



GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III

UNIT 1: MATRICES

LECTURE 2

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING



MATRICES LECTURE 2

TOPICS TO BE COVERED:

- Transpose of a matrix
- Symmetric and Skew-symmetric matrix
- Minor, Cofactor, Determinant of a square matrix
- Adjoint and Inverse of a square matrix



Transpose of a Matrix:

- Let $A = [a_{ij}]_{m \times n}$ be a matrix. The transpose of A denoted by A' or A^T is the matrix of order $n \times m$ obtained by interchanging rows and columns of A.

$$A' = [a_{ji}]_{n \times m}$$

Example: If $A = \begin{bmatrix} 2 & 4 & 6 \\ -1 & 3 & 2 \end{bmatrix}_{2 \times 3}$, then $A' = \begin{bmatrix} 2 & -1 \\ 4 & 3 \\ 6 & 2 \end{bmatrix}_{3 \times 2}$

Properties of transpose: Let A and B be two matrices conformable for the indicated operations. Then

1. $(A')' = A$
2. $(A+B)' = A' + B'$
3. $(kA') = kA'$
4. $(AB)' = B'A'$



Symmetric Matrix:

Definition: A square matrix $A = [a_{ij}]_{n \times n}$ is said to be symmetric if $A' = A$
i.e. $a_{ij} = a_{ji} \quad \forall i, j$

Example: Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & -1 \\ 3 & -1 & 5 \end{bmatrix}$

$$A' = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & -1 \\ 3 & -1 & 5 \end{bmatrix} = A$$

Therefore A is symmetric matrix.



Skew-Symmetric Matrix:

Definition: A square matrix $A = [a_{ij}]_{n \times n}$ is said to be skew-symmetric if $A' = -A$

$$\text{i.e. } a_{ij} = -a_{ji} \quad \forall i, j$$

For diagonal elements, $i = j$

$$\therefore a_{ii} = -a_{ii}$$

$$\Rightarrow a_{ii} = 0 \quad \forall i$$

i.e The diagonal elements of a skew symmetric matrix are zeroes.

$$\text{Let } B = \begin{bmatrix} 0 & 2 & -3 \\ -2 & 0 & 2 \\ 3 & -2 & 0 \end{bmatrix}$$

$$B' = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & -2 \\ -3 & 2 & 0 \end{bmatrix} = -B$$

Therefore above matrix B is skew symmetric.



Properties of symmetric and Skew-Symmetric Matrix:

1. If A and B are symmetric , then

- i. $A + B, A - B, kA$ is symmetric
- ii. AB is symmetric only if $AB = BA$

Proof: Since A and B are symmetric $A' = A$ and $B' = B$

$$(A + B)' = A' + B' = A + B. \quad \therefore A + B \text{ is symmetric}$$

$$(A - B)' = A' - B' = A - B. \quad \therefore A - B \text{ is symmetric}$$

$$(kA)' = kA' = kA. \quad \therefore kA \text{ is symmetric}$$

$$(AB)' = B'A' = BA = AB \quad \therefore AB \text{ is symmetric only if } AB = BA$$

2. If A and B are skew symmetric , then

- i. $A + B, A - B, kA$ is skew symmetric
- ii. AB is symmetric only if $AB = BA$

Proof: Since A and B are skew symmetric $A' = -A$ and $B' = -B$

$$(A + B)' = A' + B' = -A - B = -(A + B) \quad \therefore A + B \text{ is skew-symmetric}$$

$$(A - B)' = A' - B' = -A + B = -(A - B) \quad \therefore A - B \text{ is skew-symmetric}$$

$$(kA)' = kA' = k(-A) = -(kA) \quad \therefore kA \text{ is skew-symmetric}$$

$$(AB)' = B'A' = (-B)(-A) = BA = AB \quad \therefore AB \text{ is symmetric only if } AB = BA$$

3. If A is any square matrix then $A + A'$ is symmetric and $A - A'$ is Skew-Symmetric



Important Theorem:

Theorem : Every square matrix A can be uniquely expressed as sum of symmetric and skew symmetric matrix.

Proof: Let A be any square matrix ,

$$\begin{aligned} A &= \frac{1}{2} [A + A'] + \frac{1}{2} [A - A'] \\ &= P + Q \quad \text{where } P = \frac{1}{2} [A + A'] \text{ and } Q = \frac{1}{2} [A - A'] \end{aligned}$$

To prove P is symmetric and Q is skew symmetric

$$\text{Consider } P' = \left[\frac{1}{2} (A + A') \right]' = \frac{1}{2} (A + A')' = \frac{1}{2} [A' + (A')'] = \frac{1}{2} [A' + A] = \frac{1}{2} [A + A'] = P$$

Therefore P is symmetric.

$$\text{Now consider } Q' = \left[\frac{1}{2} (A - A') \right]' = \frac{1}{2} (A - A')' = \frac{1}{2} [A' - (A')'] = \frac{1}{2} [A' - A] = -\frac{1}{2} [A - A'] = -Q$$

Therefore Q is Skew-symmetric.



Important Theorem:

Uniqueness : Let us assume that $A = R + S$, where R is symmetric and S is skew symmetric
Then,

$$A' = [R + S]' = R' + S' = R - S$$

$$A + A' = R + S + R - S = 2R$$

$$A - A' = R + S - (R - S) = 2S$$

Solving the above two equations $A + A' = 2R$ and $A - A' = 2S$

$$\text{we get } R = \frac{1}{2} [A + A'] = P$$

$$\text{and } S = \frac{1}{2} [A - A'] = Q$$

Therefore P and Q are unique.



Determinant of a matrix

Every *square* matrix A can be associated with a value called its **determinant** denoted by $|A|$ or $\det(A)$.

1. If $A = [a_{11}]_{1 \times 1}$, then $|A| = a_{11}$

2. If $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ then $|A|$
 $= a_{11} \cdot a_{22} - a_{21} \cdot a_{12}$

3. If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ then $|A|$
 $= a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$

Note: A square matrix A is said to be **singular** if $|A| = 0$, **non-singular** if $|A| \neq 0$



Determinant continued

Example 1. If $A = [4]_{1 \times 1}$, then $|A| = 4$

Example 2. If $A = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$ then $|A| = 4 \cdot 2 - 3 \cdot 1 = 5$

Example 3. If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 3 \\ -1 & 0 & 1 \end{bmatrix}$ then $|A| = (1)(2 - 0) - (0)(0 + 3) + (1)(0 + 2) = 4$



Properties of Determinant

- If A and B are square matrices of the order n then
 1. $|A| = |A'|$ i.e a square matrix and its transpose have the same determinant.
 2. If two rows/columns of a square matrix are proportional then $|A| = 0$.
 3. If any row/column of a square matrix A is zero then $|A| = 0$.
 4. If the elements of any row /column are all multiplied by a non zero number then its determinant is multiplied by that number. This operation is denoted by $R_i \rightarrow kR_i$. Thus $|kA| = k^n |A|$. Thus $|-A| = (-1)^n |A|$.
 5. Interchanging any two rows/columns of a square matrix changes the determinant by sign only. Interchanging row R_i and R_j is denoted by $R_i \leftrightarrow R_j$ and $C_i \leftrightarrow C_j$ for columns.
 6. $|AB| = |A| |B|$.
 7. Multiplying a constant to any row/column and adding to another row/column does not alter the determinant. Multiplying k to R_j and adding to R_i is denoted $R_i \rightarrow R_i + kR_j$.
 8. The determinant of a diagonal, upper/lower triangular matrix is the product of its diagonal elements.



Minor and Cofactor

Definition: If A is a square matrix, then

- The Minor of element a_{ij} , denoted by m_{ij} is defined to be the determinant of the submatrix obtained after deleting the i th row and j th column from A .
- The number $(-1)^{i+j} m_{ij}$ denoted by C_{ij} is called the cofactor of element a_{ij}

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

The minor of a_{11} is $m_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

The minor of a_{12} is $m_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

The minor of a_{13} is $m_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

$$C_{ij} = (-1)^{i+j} m_{ij}$$

$$C_{11} = (-1)^{1+1} m_{11} = +m_{11}$$

$$C_{12} = (-1)^{1+2} m_{12} = -m_{12}$$

$$C_{13} = (-1)^{1+3} m_{13} = +m_{13}$$

$$\begin{bmatrix} + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ + & - & + & - & + & \dots \\ - & + & - & + & - & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$



Adjoint and Cofactor matrix

Expansions by Cofactors: The determinant of an $n \times n$ matrix A can be computed by multiplying the entries in any row (or column) by their cofactors and adding the resulting products; that is, for each $1 \leq i, j \leq n$

$$|A| = a_{i1}c_{i1} + a_{i2}c_{i2} \dots + a_{in}c_{in} \quad (\text{cofactor expansion along the } i^{\text{th}} \text{ row})$$

$$|A| = a_{1j}c_{1j} + a_{2j}c_{2j} + \dots + a_{nj}c_{nj} \quad (\text{cofactor expansion along the } j^{\text{th}} \text{ column})$$

Note: $a_{i1}c_{j1} + a_{i2}c_{j2} \dots + a_{in}c_{jn} = 0$ if $i \neq j$

Definition : The cofactor matrix of $\mathbf{A}$ is the square matrix of the same order as $\mathbf{A}$ in which each element a_{ij} is replaced by its cofactor c_{ij} .

$$\begin{aligned} \text{cof } A \\ = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} \end{aligned}$$

Definition: The adjoint matrix of $\mathbf{A}$, denoted by $\text{adj}(\mathbf{A})$, is the transpose of its cofactor matrix.
i.e. $\text{adj}(A) = (\text{cof } A)'$



Cofactors & Adjoint continued

Let A

$$= \begin{bmatrix} 3 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

The cofactors are

$$c_{11} = -1$$

$$c_{21} = 1$$

$$c_{31} = -1$$

$$c_{12} = 2$$

$$c_{22} = -4$$

$$c_{32} = 2$$

$$c_{13} = 3$$

$$c_{23} = -7$$

$$c_{33} = 5$$

$\text{Cof } A$

$$= \begin{bmatrix} -1 & 2 & 3 \\ 1 & -4 & -7 \\ -1 & 2 & 5 \end{bmatrix}$$

$\text{adj } A = (\text{cof } A)'$

$$= \begin{bmatrix} -1 & 1 & -1 \\ 2 & -4 & 2 \\ 3 & -7 & 5 \end{bmatrix}$$

Note: $A(\text{adj } A) = (\text{adj } A) A = |A| I$



Inverse of a matrix

Definition: Let $A = [a_{ij}]_{n \times n}$. If there exists a matrix $B = [b_{ij}]_{n \times n}$ such that $AB = I_n = BA$, then B is called inverse of A and is denoted by A^{-1} .

Note: A^{-1} exists if $|A| \neq 0$ (non singular)

$$A^{-1} = \frac{1}{|A|} \text{adj } A$$

$$\text{since } A (\text{adj } A) = (\text{adj } A) A = |A| I$$

The inverse, if it exists is unique

Example 1:

$$\text{Let } A = \begin{bmatrix} 1 & 2 \\ -3 & 4 \end{bmatrix}; |A| = 10$$

$$A^{-1} = \frac{1}{10} \begin{bmatrix} 4 & -2 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} 0.4 & -0.2 \\ 0.3 & 0.1 \end{bmatrix}$$



Solved Problem:

Let A

Example 2:
$$= \begin{bmatrix} 3 & -1 & 1 \\ 2 & 1 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

$|A| = (3)(-1-0) - (-1)(-2-0) + (1)(4-1) = -2 \neq 0$. Therefore A^{-1} exists

$$\text{cof } A = \begin{bmatrix} -1 & 2 & 3 \\ 1 & -4 & -7 \\ -1 & 2 & 5 \end{bmatrix}$$

$$\begin{aligned} \text{adj } A &= (\text{cof } A)' \\ &= \begin{bmatrix} -1 & 1 & -1 \\ 2 & -4 & 2 \\ 3 & -7 & 5 \end{bmatrix} \end{aligned}$$

$$A^{-1} = \frac{\text{adj } A}{|A|} = \frac{1}{-2} \begin{bmatrix} -1 & 1 & -1 \\ 2 & -4 & 2 \\ 3 & -7 & 5 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 & 0.5 \\ -1.0 & 2.0 & -1.0 \\ -1.5 & 3.5 & -2.5 \end{bmatrix}$$



EXCERCISE

- Without expanding show that (Specify the operations)

$$\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0, \quad \begin{vmatrix} 1/a & a^2 & bc \\ 1/b & b^2 & ca \\ 1/c & c^2 & ab \end{vmatrix} = 0,$$

$$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$$



Excercise

- Show that a skew-symmetric matrix of odd order is singular.

- Write the matrix $\begin{bmatrix} 1 & 5 & 3 \\ 1 & 6 & -7 \\ 2 & 3 & 4 \end{bmatrix}$ as a sum of symmetric and skew-symmetric matrix.

- Find the adjoint and inverse of the matrix $\begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$.



Excercise

- Find the condition on k so that the matrix $\begin{bmatrix} 1 & 3 & 4 \\ 3 & k & 6 \\ -1 & 5 & 1 \end{bmatrix}$ has an inverse.
- If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ then show that $A^n = \begin{bmatrix} 1 + 2n & -4n \\ n & 1 - 2n \end{bmatrix}$. Use the principal of mathematical induction.

THANK YOU!

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Solution of Exercise

$$\begin{aligned}
 & \bullet \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & b+c+a \\ 1 & c & c+a+b \end{vmatrix} \quad C_3 \rightarrow C_3 + C_2 \\
 & \bullet = (a+b+c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} = 0 \quad \text{since two columns are identical.} \\
 & \bullet \begin{vmatrix} 1/a & a^2 & bc \\ 1/b & b^2 & ca \\ 1/c & c^2 & ab \end{vmatrix} = \frac{1}{abc} \begin{vmatrix} 1 & a^3 & abc \\ 1 & b^3 & bca \\ 1 & c^3 & cab \end{vmatrix} \quad R_1 \rightarrow aR_1, R_2 \rightarrow bR_2, R_3 \rightarrow cR_3
 \end{aligned}$$

Solution of Exercise

- $$\begin{vmatrix} 1 & a^3 & abc \\ 1 & b^3 & bca \\ 1 & c^3 & cab \end{vmatrix} = abc \begin{vmatrix} 1 & a^3 & 1 \\ 1 & b^3 & 1 \\ 1 & c^3 & 1 \end{vmatrix} = 0$$

since two columns are identical

- $$\begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = abc \begin{vmatrix} 1/a & 1 & a \\ 1/b & 1 & b \\ 1/c & 1 & c \end{vmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{a}R_1, R_2 \rightarrow \frac{1}{b}R_2, R_3 \rightarrow \frac{1}{c}R_3}$$

Solution of Exercise

$$\begin{aligned}
 & \bullet \text{ abc } \begin{vmatrix} 1/a & 1 & a \\ 1/b & 1 & b \\ 1/c & 1 & c \end{vmatrix} = \begin{vmatrix} bc & 1 & a \\ ac & 1 & b \\ ab & 1 & c \end{vmatrix} C_1 \longrightarrow \text{abc} C_1 \\
 & \bullet = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} C_1 \longleftrightarrow C_2, C_2 \longleftrightarrow C_3.
 \end{aligned}$$

Solution of Exercise

- Find the adjoint and inverse of the matrix $\begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$.

Type equation here.

- Cofactor matrix =
$$\begin{bmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} & - \begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} & + \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} \\ - \begin{vmatrix} 5 & 3 \\ 2 & 1 \end{vmatrix} & + \begin{vmatrix} 2 & 3 \\ 1 & 1 \end{vmatrix} & - \begin{vmatrix} 2 & 5 \\ 1 & 2 \end{vmatrix} \\ + \begin{vmatrix} 5 & 3 \\ 1 & 2 \end{vmatrix} & - \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} & + \begin{vmatrix} 2 & 5 \\ 3 & 1 \end{vmatrix} \end{bmatrix}$$

Solution of Exercise

- Cofactor matrix=A=
$$\begin{bmatrix} +(-3) & -(1) & +(5) \\ -(-1) & +(-1) & -(-1) \\ +(7) & -(-5) & +(-13) \end{bmatrix} =$$

$$A = \begin{bmatrix} -3 & -1 & 5 \\ 1 & -1 & 1 \\ 7 & 5 & -13 \end{bmatrix}$$

- Adjoint matrix=A'=
$$\begin{bmatrix} -3 & 1 & 7 \\ -1 & -1 & 5 \\ 5 & 1 & -13 \end{bmatrix}$$

Solution of Exercise

- $\begin{vmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{vmatrix} = 2(1-4) - 5(3-2) + 3(6-1) = 2 \times -3 - 5 \times 1 + 3 \times 5 = -6 - 5 + 15 = 4$

- $\text{Inverse} = \frac{1}{4} \text{adjoint matrix} = \frac{1}{4} \begin{bmatrix} -3 & 1 & 7 \\ -1 & -1 & 5 \\ 5 & 1 & -13 \end{bmatrix}$

Solution of Exercise

If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ then show that $A^n = \begin{bmatrix} 1 + 2n & -4n \\ n & 1 - 2n \end{bmatrix}$. Use the principal of mathematical induction.

Sol: For $n=1$ it is clearly satisfied.

Assume for $n=k$, $A^k = \begin{bmatrix} 1 + 2k & -4k \\ k & 1 - 2k \end{bmatrix}$

To show that it is satisfied for $n=k+1$.

$$A^{k+1} = A^k A = \begin{bmatrix} 1 + 2k & -4k \\ k & 1 - 2k \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 + 2k & -4 - 4k \\ k + 1 & -1 - 2k \end{bmatrix}$$

Solution of Exercise

$$A^{k+1} = \begin{bmatrix} 3 + 2k & -4 - 4k \\ k + 1 & -1 - 2k \end{bmatrix} = \begin{bmatrix} 1 + 2(k + 1) & -4(k + 1) \\ k + 1 & 1 - 2(k + 1) \end{bmatrix}$$

Therefore by the principal of mathematical induction

$$A^n = \begin{bmatrix} 1 + 2n & -4n \\ n & 1 - 2n \end{bmatrix}$$

Application of matrices to Cryptography

- In cryptography a message is encoded using square matrix called encoding matrix and decoded by decoding matrix which is the inverse of the encoding matrix.
- It starts with assigning all the element of the source code an integer. Suppose the alphabets A,B,.....Z are assigned 1,2,3,..26 and space is assigned 0. A message is converted to its integer representation.
- Then by using the encoding matrix it is converted to a code. This code is transmitted. The receiver use the decoding matrix to retrieve the original message.
- Suppose password 'AKMDBS' has to conveyed by coding. The message converting to integers will be [1 11 13 4 2 19]. This is written in the form of matrix

Application of matrices to Cryptography

$$X = \begin{bmatrix} 1 & 11 & 13 \\ 4 & 2 & 19 \end{bmatrix}.$$

- We choose a non-singular matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ as the encoding matrix. The code generated is XA which is

- $XA = \begin{bmatrix} 1 & 11 & 13 \\ 4 & 2 & 19 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 14 & 26 & 13 \\ 23 & 29 & 19 \end{bmatrix} = Y$

Application of Matrices to Cryptography

- The code [14 26 13 23 29 19] is transmitted. The receiver is provided the decoding matrix, which is inverse of A and is

$$\begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

- Since $Y=XA$, we obtain by multiplying A^{-1} on the right we get

$$Y A^{-1} = X A A^{-1} = X.$$

$$\text{There for } X=Y A^{-1} = \begin{bmatrix} 14 & 26 & 13 \\ 23 & 29 & 19 \end{bmatrix} \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 11 & 13 \\ 4 & 2 & 19 \end{bmatrix}$$

Application of matrices to Cryptography

- The code $\begin{bmatrix} 1 & 11 & 13 \\ 4 & 2 & 19 \end{bmatrix}$ which $[1 \ 11 \ 13 \ 4 \ 2 \ 19]$ is translated to alphabets AKMDBS.

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Application of Matrices to Geometry

- Consider cartesian coordinate system with coordinate axis X and Y and a point P(x,y). Rotate the coordinate axis by angle θ and let the new coordinate axis be X' and Y'. Let the coordinates of P in the new coordinate system be (x',y'). The relation between the (x,y) and (x',y') can be shown to be $x' = x \cos \theta - y \sin \theta$; $y' = x \sin \theta + y \cos \theta$ which can be written as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

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